In [1]:
```
using BifurcationKit, LinearAlgebra, Parameters
using JLD2
#using GLMakie
using ForwardDiff
using Plots
theme(:dark)
using DelimitedFiles
const BK = BifurcationKit
#using ProgressMeter
using JupyterFormatter
enable_autoformat();
```

In [2]:
```
function tf(x)
    return exp.(x) ./ (1.0 .+ exp.(x))
end
```

Out[2]: tf (generic function with 1 method)

In [3]:
```
function itf(y)
    return log.(y ./ (1 .- y))
end
```

Out[3]: itf (generic function with 1 method)

In [4]:
```
function fe(phi, p)
    @unpack d2, chi, N, nuA, lambda, switch_dh, zplus, zpep = p

    phiplus = phi[1]
    phiminus = phi[2]
    phipep = phi[3]
    phioh = phi[4]
    phiw = phi[5]

    fFH = (
        (phiplus > 0 ? phiplus * log(phiplus) : 0) +
        (phiminus > 0 ? phiminus * log(phiminus) : 0) +
        (phipep > 0 ? phipep * log(phipep) / (N * nuA) : 0) +
        (phioh > 0 ? phioh * log(phioh) : 0) +
        (phiw > 0 ? phiw * log(phiw) : 0) +
        chi * phipep * phiw
    )


    kappa = sqrt(
        lambda * (
            zplus * zplus * abs(phiplus) +
            abs(phiminus) +
            abs(phioh) +
            (zpep[1]^2 + zpep[2]^2) / (N * nuA) * abs(phipep)
        ),
    )

    fDH = -1 / (4 * pi) * (log(1 + kappa) - kappa + kappa^2 / 2)

    return fFH + switch_dh * fDH
end
```

Out[4]:
```
fe (generic function with 1 method)
```

In [5]:
```
gradfe(x, p) = ForwardDiff.gradient(y -> fe(y, p), x)
```

Out[5]:
```
gradfe (generic function with 1 method)
```

In [6]:
```
Hfe(x, p) = ForwardDiff.hessian(y -> fe(y, p), x)
```

Out[6]:
```
Hfe (generic function with 1 method)
```

In [7]:
```
function evmin(y, p)
    @unpack d2, chi, N, nuA, lambda, zplus, zpep = p

    phiplus = y[1]
    phiminus = y[2]
    phipep = y[3]
    phioh = (zpep[1] + zpep[2]) / (N * nuA) * phipep
    phiw = 1 - phiplus - phiminus - phipep - phioh

    x = [phiplus, phiminus, phipep, phioh, phiw]

    Hall = Hfe(x, p)

    G = [
        1 0 0 zplus -(1 + zplus)
        0 1 0 -1 0
        0 0 1 (zpep[1]+zpep[2])/(N*nuA) -(1 + (zpep[1] + zpep[2]) / (N *
    ]

    H = G * Hall * G'

    ev = eigvals(H)

    return ev[argmin(abs.(ev))]
end
```

Out[7]:
```
evmin (generic function with 1 method)
```

In [8]:
```
function F_dhlen(x, p)
    f = similar(x)
    n = length(x)

    @unpack d2, chi, N, nuA, lambda, zplus, KM, zpep = p

    phi1_1 = tf(x[1])
    phi2_1 = tf(x[2])
    phi3_1 = tf(x[3])
    phi4_1 = tf(x[4])
    phi5_1 = tf(x[5])

    phi1_2 = tf(x[6])
    phi2_2 = tf(x[7])
    phi3_2 = tf(x[8])
    phi4_2 = tf(x[9])
    phi5_2 = tf(x[10])

    lamnv1 = x[11]
    lamnv2 = x[12]
```

```
    lamen1 = x[13]
    lamen2 = x[14]

    om1 = x[15]
    om2 = x[16]
    om3 = x[17]


    fe_1 = fe([phi1_1, phi2_1, phi3_1, phi4_1, phi5_1], p)
    fe_2 = fe([phi1_2, phi2_2, phi3_2, phi4_2, phi5_2], p)

    gfe1_1, gfe2_1, gfe3_1, gfe4_1, gfe5_1 =
        gradfe([phi1_1, phi2_1, phi3_1, phi4_1, phi5_1], p)
    gfe1_2, gfe2_2, gfe3_2, gfe4_2, gfe5_2 =
        gradfe([phi1_2, phi2_2, phi3_2, phi4_2, phi5_2], p)

    a = [zplus, -1, (zpep[1] + zpep[2]) / (N * nuA), -1, 0]

    f[1] = gfe1_1 + lamnv1 + lamen1 * a[1] + om1
    f[2] = gfe2_1 + lamnv1 + lamen1 * a[2] + om2
    f[3] = gfe3_1 + lamnv1 + lamen1 * a[3] + om3
    f[4] = gfe4_1 + lamnv1 + lamen1 * a[4]
    f[5] = gfe5_1 + lamnv1 + lamen1 * a[5]

    f[5+1] = gfe1_2 + lamnv2 + lamen2 * a[1] + om1
    f[5+2] = gfe2_2 + lamnv2 + lamen2 * a[2] + om2
    f[5+3] = gfe3_2 + lamnv2 + lamen2 * a[3] + om3
    f[5+4] = gfe4_2 + lamnv2 + lamen2 * a[4]
    f[5+5] = gfe5_2 + lamnv2 + lamen2 * a[5]

    f[11] =
        fe_1 - fe_2 +
        om1 * (phi1_1 - phi1_2) +
        om2 * (phi2_1 - phi2_2) +
        om3 * (phi3_1 - phi3_2)

    f[12] = phi1_1 + phi2_1 + phi3_1 + phi4_1 + phi5_1 - 1
    f[13] = phi1_2 + phi2_2 + phi3_2 + phi4_2 + phi5_2 - 1
    f[14] = a[1] * phi1_1 + a[2] * phi2_1 + a[3] * phi3_1 + a[4] * phi4_1
    f[15] = a[1] * phi1_2 + a[2] * phi2_2 + a[3] * phi3_2 + a[4] * phi4_2

    f[16] = phi3_1 - KM
    f[17] = zplus * phi1_1 - phi2_1

    return f

end
```

Out[8]: F_dhlen (generic function with 1 method)

In [9]:
```
function build_run(case)

    phi1_1 = case.phivar1[2]
    phi2_1 = case.phivar1[3]
    phi3_1 = case.phivar1[1]
    phi4_1 =
        case.zplus * phi1_1 - phi2_1 +
        (case.zpep[1] + case.zpep[2]) / (case.N * case.nuA) * phi3_1
    phi5_1 = 1 - phi1_1 - phi2_1 - phi3_1 - phi4_1
```

```
    phi1_2 = case.phivar2[2]
    phi2_2 = case.phivar2[3]
    phi3_2 = case.phivar2[1]
    phi4_2 =
        case.zplus * phi1_2 - phi2_2 +
        (case.zpep[1] + case.zpep[2]) / (case.N * case.nuA) * phi3_2
    phi5_2 = 1 - phi1_2 - phi2_2 - phi3_2 - phi4_2

    dd2 = (phi1_1 - phi1_2)^2 + (phi2_1 - phi2_2)^2 + (phi3_1 - phi3_2)^2

    par = (
        d2 = dd2,
        chi = case.chi,
        N = case.N,
        nuA = case.nuA,
        lambda = case.lambda,
        KM = case.KM,
        switch_dh = case.switch_dh,
        zplus = case.zplus,
        zpep = case.zpep,
        p_max = case.p_max,
        p_min = case.p_min,
    )

    gfe1_1, gfe2_1, gfe3_1, gfe4_1, gfe5_1 =
        gradfe([phi1_1, phi2_1, phi3_1, phi4_1, phi5_1], par)
    gfe1_2, gfe2_2, gfe3_2, gfe4_2, gfe5_2 =
        gradfe([phi1_2, phi2_2, phi3_2, phi4_2, phi5_2], par)

    #    a = [zplus , -1, 2 / (N * nuA), -1, 0]


    lamnv1 = -gfe5_1
    lamnv2 = -gfe5_2
    lamen1 = lamnv1 + gfe4_1
    lamen2 = lamnv2 + gfe4_2
    om1 = -(gfe1_1 + lamnv1 + lamen1 * case.zplus)
    om2 = -(gfe2_1 + lamnv1 + lamen1 * (-1))
    # CHECK!!
    om3 = -(gfe3_1 + lamnv1 + lamen1 * (case.zpep[1] + case.zpep[2]) / (c

    sol0 = [
        itf(phi1_1),
        itf(phi2_1),
        itf(phi3_1),
        itf(phi4_1),
        itf(phi5_1),
        itf(phi1_2),
        itf(phi2_2),
        itf(phi3_2),
        itf(phi4_2),
        itf(phi5_2),
        lamnv1,
        lamnv2,
        lamen1,
        lamen2,
        om1,
        om2,
        om3,
    ]
```

```
    return (sol0 = sol0, par = par, phivar1 = case.phivar1, phivar2 = cas
end
```

Out[9]: build_run (generic function with 1 method)

In [10]:
```
# Case 1 should correspond to the violet line on fig 1(a) in my and Giuli
case1 = (
    chi = 0.85,
    N = 7,
    nuA = 50 / 7,
    lambda = 100.0,
    #    KM = 1.35e-3,
    KM = 1.5 * 1e-3 * 50 / 55.35,
    switch_dh = 1,
    zplus = 1.0,
    zpep = [1, 1],
    p_max = 125.0,
    p_min = 0.1,
    phivar1 = (
        0.0014072188552725978,
        0.013975575489771453,
        0.013975575489771453,
        5.420054200542007e-5,
        0,
        1.9049897537435612,
    ),
    phivar2 = (
        0.4713311048662077,
        0.00752500303794388,
        0.026272431023913592,
        9.901500213468598e-5,
        0.5717491375633551,
        2.5747871316756075,
    ),
);

# Case 2 ROUGHLY corresponds to the violet line on fig 1(b) in my and Giu
case2 = (
    chi = 0.8555,
    N = 7,
    nuA = 50 / 7,
    lambda = 120.0,
    #    KM = 1.35e-3,
    KM = 1.5 * 1e-3 * 50 / 55.35,
    switch_dh = 1,
    zplus = 2.0,
    zpep = [1, 1],
    p_max = 124.0,
    p_min = 0.1,
    phivar1 = (
        0.0013550135501355016,
        1.3089264695931858e-5,
        2.6178528880041418e-5,
        6.855740370939678e-5,
        0,
        0.1410653171572616,
        -0.5902705704964453,
    ),
    phivar2 = (
```

```
        0.497835398462283,
        0.0008572932662572949,
        0.007043988949735419,
        0.01991030492217057,
        3.2482754058030023,
        2.2923143560178105,
        -3.3244075877556547,
    ),
);
```

In [11]:
```
ACTIVE_CASE = case2
cfg = build_run(ACTIVE_CASE)

sol0 = cfg.sol0
par = cfg.par
dd2 = par.d2
phivar1 = cfg.phivar1
phivar2 = cfg.phivar2

p_max = par.p_max
p_min = par.p_min

@show sol0;
@show phivar1;
@show phivar2;
@show par

#@show F_dhlen(sol0, par);
```

```
sol0 = [-11.243705063231525, -10.550544812701622, -6.602587892189336, -9.8
22765438014132, 6.535789714075295, -7.060872835951271, -4.948511742136696,
-0.008658460243423606, -4.213137908811369, -0.08132757339727377, -0.999709
6821419033, -0.6912606721016679, -10.444495245154036, -7.256902773814488,
34.62028201697818, 0.7277513515476137, 0.7001861816132058]
phivar1 = (0.0013550135501355016, 1.3089264695931858e-5, 2.617852888004141
8e-5, 6.855740370939678e-5, 0, 0.1410653171572616, -0.5902705704964453)
phivar2 = (0.497835398462283, 0.0008572932662572949, 0.007043988949735419,
0.01991030492217057, 3.2482754058030023, 2.2923143560178105, -3.3244075877
556547)
par = (d2 = 0.24654273494601342, chi = 0.8555, N = 7, nuA = 7.142857142857
143, lambda = 120.0, KM = 0.0013550135501355014, switch_dh = 1, zplus = 2.
0, zpep = [1, 1], p_max = 124.0, p_min = 0.1)
```

Out[11]:
```
(d2 = 0.24654273494601342, chi = 0.8555, N = 7, nuA = 7.142857142857143,
lambda = 120.0, KM = 0.0013550135501355014, switch_dh = 1, zplus = 2.0,
zpep = [1, 1], p_max = 124.0, p_min = 0.1)
```

In [12]:
```
# qdlen = plot()

# ev0(phiplus,phipep)=evmin(phiplus,phiminus/par.zplus,phipep)

phiplus_r = range(0.00001, 0.4 + 0.0 / (1 + par.zplus), length = 1000)
phipep_r = range(
    0.00001,
    0.5 + 0.0 / (1 + (par.zpep[1] + par.zpep[2]) / (par.N * par.nuA)),
    length = 1000,
)

Z = [evmin([x, x * par.zplus, y], par) for y in phipep_r, x in phiplus_r]

cp =
```

```
    contour(phiplus_r, phipep_r, Z; levels = [0], linewidth = 2, color =

# Zdet = [hessdet([x, x * par.zplus, y], par) for y in phipep_r, x in phi

cpdet = deepcopy(cp)

# contour!(
#     cpdet,
#     phiplus_r,
#     phipep_r,
#     Zdet;
#     levels = [0],
#     linewidth = 2,
#     color = :white,
#     linestyle = :dash,
#     cbar = false,
# )

display(cpdet)

# plot!(
#     qdlen,
#     br,
#     vars = (:param, :phiplus1),
#     branchlabel = ["br"],
#     legend = :topright,
#     color = :red,
# )
# display(qdlen)

# using DataFrames, CSV

# df = DataFrame(
#     phiplus = repeat(phiplus_r, outer = length(phipep_r)),
#     phipep = repeat(phipep_r, inner = length(phiplus_r)),
#     Z = vec(Z),
# )

# CSV.write("contour_data.csv", df)


# using Printf
# import Contour

# Zc = permutedims(Z)  # convert from rows=y, cols=x to rows=x, cols=y

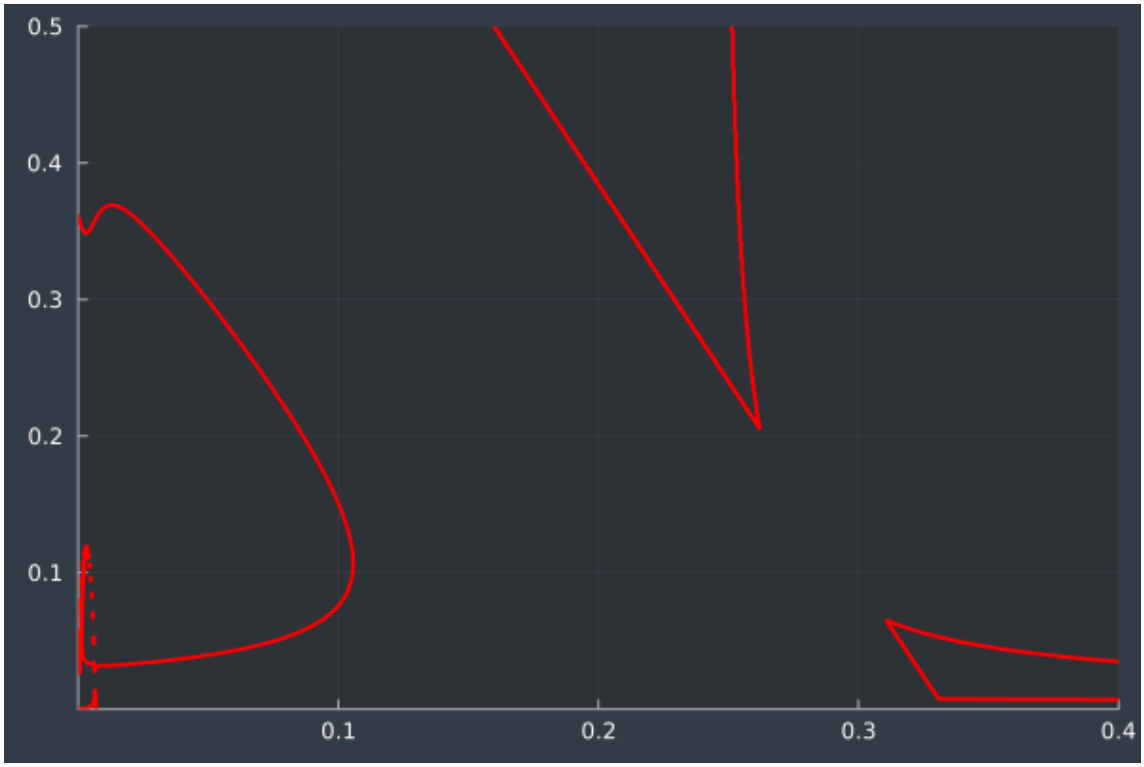
# open("contour0.dat", "w") do io
#     println(io, "# contour data")
#     println(io, "# columns: x y")
#     println(io, "# blank line separates different contour lines")

#     cs = Contour.contours(phiplus_r, phipep_r, Zc, [0.0])

#     seg = 0
#     for cl in Contour.levels(cs)
#         for ln in Contour.lines(cl)
#             seg += 1
#             xs, ys = Contour.coordinates(ln)

#             println(io, "# segment ", seg)
```

```
#             for (x, y) in zip(xs, ys)
#                 @printf(io, "%.16g %.16g\n", x, y)
#             end
#             println(io)  # empty line between contour lines
#         end
#     end
# end
```



In [13]:
```
probdlen = BK.BifurcationProblem(
    F_dhlen,
    sol0,
    par,
    (@optic _.lambda),
    record_from_solution = (x, p; kwargs...) -> (
        phiplus1 = tf(x[1]),
        phiminus1 = tf(x[2]),
        phipep1 = tf(x[3]),
        phioh1 = tf(x[4]),
        phiw1 = tf(x[5]),
        phiplus2 = tf(x[5+1]),
        phiminus2 = tf(x[5+2]),
        phipep2 = tf(x[5+3]),
        phioh2 = tf(x[5+4]),
        phiw2 = tf(x[5+5]),
        delphipm1 = tf(x[2]) - tf(x[1]),
        delphipm2 = tf(x[5+2]) - tf(x[5+1]),
    ),
)

#optnewton = NewtonPar(tol = 1e-8, verbose = true, linesearch = true, α =
optnewton = NewtonPar(tol = 1e-8, verbose = true, linesearch = true, α =
sol = BK.solve(probdlen, Newton(), optnewton)


println(
    "phiplus1 = $(tf(sol.u[1]))\n" *
    "phiminus1 = $(tf(sol.u[2]))\n" *
```

```
    "phipep1 = $(tf(sol.u[3]))\n" *
    "phioh1 = $(tf(sol.u[4]))\n" *
    "phiw1 = $(tf(sol.u[5]))\n" *
    "phiplus2 = $(tf(sol.u[5+1]))\n" *
    "phiminus2 = $(tf(sol.u[5+2]))\n" *
    "phipep2 = $(tf(sol.u[5+3]))\n" *
    "phioh2 = $(tf(sol.u[5+4]))\n" *
    "phiw2 = $(tf(sol.u[5+5]))\n",
)

println(
    "phivar1 = ( \n$(tf(sol.u[3])),\n" *
    "$(tf(sol.u[1])),\n" *
    "$(tf(sol.u[2])),\n" *
    "0,\n" *
    "0,\n" *
    "0,\n" *
    "),\n" *
    "phivar2 = ( \n$(tf(sol.u[3+5])),\n" *
    "$(tf(sol.u[1+5])),\n" *
    "$(tf(sol.u[2+5])),\n" *
    "0,\n" *
    "0,\n" *
    "0,\n" *
    "),\n",
)

@show sol.prob.params.lambda;
```

Newton step	residual	linear iterations
0	2.7481e-03	0
1	5.4627e-03	1
2	7.3585e-06	1
3	2.3922e-10	1

```
phiplus1 = 1.4941534658526328e-5
phiminus1 = 2.9883069317051656e-5
phipep1 = 0.0013550135501355016
phioh1 = 5.420054200542005e-5
phiw1 = 0.9985459613039022
phiplus2 = 0.0010068845796330577
phiminus2 = 0.007755304809459192
phipep2 = 0.4951938048841877
phioh2 = 0.014066216546608208
phiw2 = 0.4819777891832686

phivar1 = (
0.0013550135501355016,
1.4941534658526328e-5,
2.9883069317051656e-5,
0,
0,
0,
),
phivar2 = (
0.4951938048841877,
0.0010068845796330577,
0.007755304809459192,
0,
0,
0,
),

sol.prob.params.lambda = 120.0
```

In [14]:
```
optscontdlen = ContinuationPar(
    dsmin = 0.000001,
    dsmax = 0.01,
    ds = 0.01,
    #        p_max = dd2*1.053499, #6/(1+par.N*par.nuA/2),
    #        p_min = dd2*0.999, #0.00001,
    p_max = p_max,
    p_min = p_min,
    nev = 0,
    detect_fold = true,
    plot_every_step = 1,
    newton_options = NewtonPar(max_iterations = 500, tol = 1e-12, verbose
    max_steps = 1000000,
    detect_bifurcation = 0,
    tol_stability = 1e-14,
)

alg = PALC(tangent = Bordered())
br = @time continuation(
    probdlen,
    alg,
    optscontdlen;
```

```
    plot = false,
    verbosity = 0,
    bothside = true,
)
```

```
 9.288207 seconds (40.06 M allocations: 2.951 GiB, 8.33% gc time, 89.43%
compilation time)
```

Out[14]:

```
┌─ Curve type: EquilibriumCont
├─ Number of points: 10614
├─ Type of vectors: Vector{Float64}
├─ Parameter lambda starts at 0.1, ends at 124.0
├─ Algo: PALC [Bordered]
└─ Special points:

- #  1, endpoint at lambda ≈ +0.10000000,
step =   0
- #  2,     fold at lambda ≈ +118.19565141 ∈ (+118.22714748, +118.227147
48), |δp|=-1e+00, [    guess], δ = ( 0,  0), step = 8359
- #  3,     fold at lambda ≈ +105.65924667 ∈ (+105.65412548, +105.654125
48), |δp|=-1e+00, [    guess], δ = ( 0,  0), step = 9265
- #  4, endpoint at lambda ≈ +124.00000000,
step = 10613
```

In [15]:

```
@show br
@show typeof(br)
br.branch[1]
```

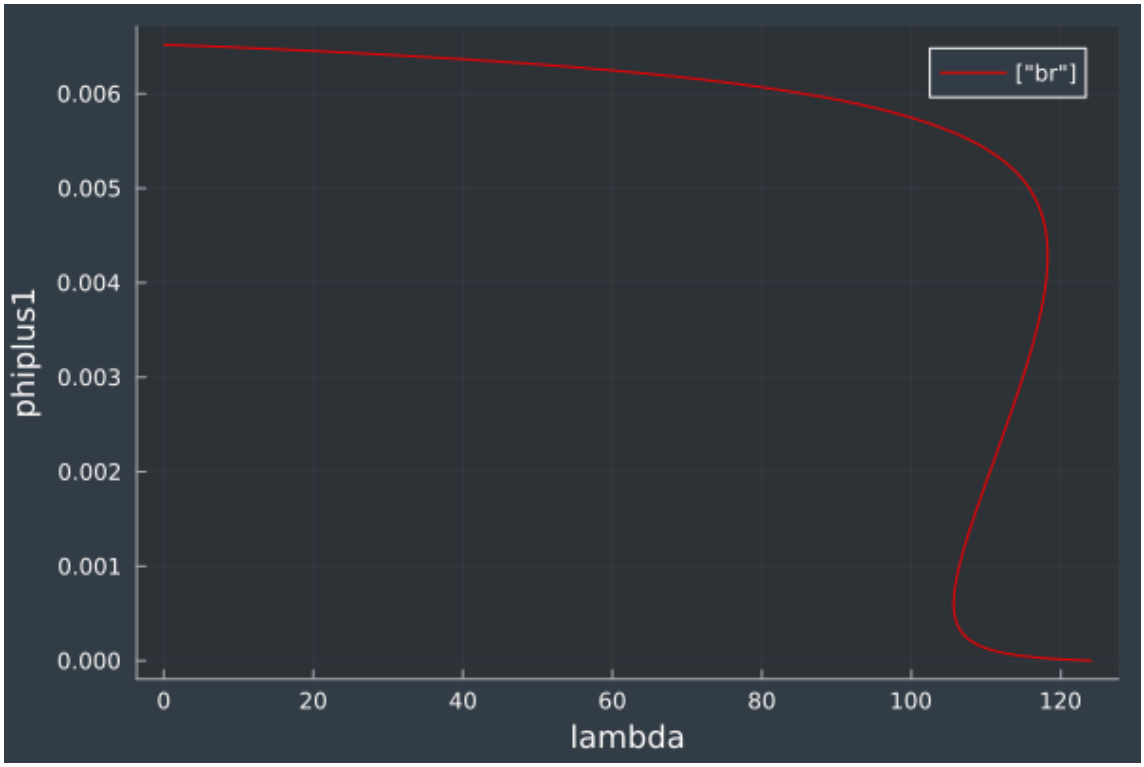
```
br =  ┌─ Curve type: EquilibriumCont
 ├─ Number of points: 10614
 ├─ Type of vectors: Vector{Float64}
 ├─ Parameter lambda starts at 0.1, ends at 124.0
 ├─ Algo: PALC [Bordered]
 └─ Special points:

- #  1, endpoint at lambda ≈ +0.10000000,
step =   0
- #  2,     fold at lambda ≈ +118.19565141 ∈ (+118.22714748, +118.2271474
8), |δp|=-1e+00, [    guess], δ = ( 0,  0), step = 8359
- #  3,     fold at lambda ≈ +105.65924667 ∈ (+105.65412548, +105.6541254
8), |δp|=-1e+00, [    guess], δ = ( 0,  0), step = 9265
- #  4, endpoint at lambda ≈ +124.00000000,
step = 10613

typeof(br) = ContResult{BifurcationKit.EquilibriumCont, @NamedTuple{phiplu
s1::Float64, phiminus1::Float64, phipep1::Float64, phioh1::Float64, phiw
1::Float64, phiplus2::Float64, phiminus2::Float64, phipep2::Float64, phioh
2::Float64, phiw2::Float64, delphipm1::Float64, delphipm2::Float64, para
m::Float64, itnewton::Int64, itlinear::Int64, ds::Float64, n_unstable::Int
64, n_imag::Int64, stable::Nothing, step::Int64}, Nothing, Nothing, Specia
lPoint{Float64, @NamedTuple{phiplus1::Float64, phiminus1::Float64, phipep
1::Float64, phioh1::Float64, phiw1::Float64, phiplus2::Float64, phiminus
2::Float64, phipep2::Float64, phioh2::Float64, phiw2::Float64, delphipm1::
Float64, delphipm2::Float64}, Vector{Float64}, Vector{Float64}}, Vector{@N
amedTuple{x::Vector{Float64}, p::Float64, step::Int64}}, ContinuationPar{F
loat64, DefaultLS, DefaultEig{typeof(real)}}, BifurcationProblem{BifFunctio
on{typeof(F_dhlen), BifurcationKit.var"#150#151"{typeof(F_dhlen)}, Nothin
g, Nothing, Nothing, Nothing, Nothing, Nothing, Nothing, Nothing, Nothing,
Float64, Nothing}, Vector{Float64}, @NamedTuple{d2::Float64, chi::Float64,
N::Int64, nuA::Float64, lambda::Float64, KM::Float64, switch_dh::Int64, zp
lus::Float64, zpep::Vector{Int64}, p_max::Float64, p_min::Float64}, Proper
tyLens{:lambda}, typeof(BifurcationKit.plot_default), var"#8#9", typeof(Bi
furcationKit.save_solution_default), typeof(BifurcationKit.update_defaul
t)}, PALC{Bordered, MatrixBLS{DefaultLS}, Float64, BifurcationKit.DotTheta
{BifurcationKit.var"#DotTheta##0#DotTheta##1", BifurcationKit.var"#DotThet
a##2#DotTheta##3"}}}
```

Out[15]:
```
(phiplus1 = 0.006516572396933145, phiminus1 = 0.013033144793866292, phip
ep1 = 0.0013550135501355016, phioh1 = 5.420054200542005e-5, phiw1 = 0.97
90410687170595, phiplus2 = 0.001143204808372582, phiminus2 = 0.021105860
09706906, phipep2 = 0.4726805694351316, phioh2 = 8.777229708136766e-5, p
hiw2 = 0.5049825933623454, delphipm1 = 0.006516572396933147, delphipm2 =
0.01996265528869648, param = 0.1, itnewton = 2, itlinear = 2, ds = -0.0
1, n_unstable = -1, n_imag = -1, stable = nothing, step = 0)
```

In [16]:
```
qdlen = plot()
plot!(
    qdlen,
    br,
    vars = (:param, :phiplus1),
    branchlabel = ["br"],
    legend = :topright,
    color = :red,
)
display(qdlen)
```



In [17]:
```
comp = 0
if (comp == 1)
    @load "mgcl2_violet_branch_data.jld2" data_pairs

    # Unpack for plotting
    params = first.(data_pairs)
    records = last.(data_pairs)

    # Plot
    plot!(
        qdlen,
        params,
        records;
        xlabel = "parameter",
        ylabel = "record",
        legend = false,
        color = :green,
        linestyle = :dash,
    )
end
```

In [18]:
```
qdlen_phiplus1 = plot()
qdlen_phiminus1 = plot()
qdlen_phipep1 = plot()
qdlen_phioh1 = plot()
qdlen_phiw1 = plot()

plot!(
    qdlen_phiplus1,
    br,
    vars = (:param, :phiminus1),
    branchlabel = ["phi-"],
    legend = :topright,
    color = :red,
)
```

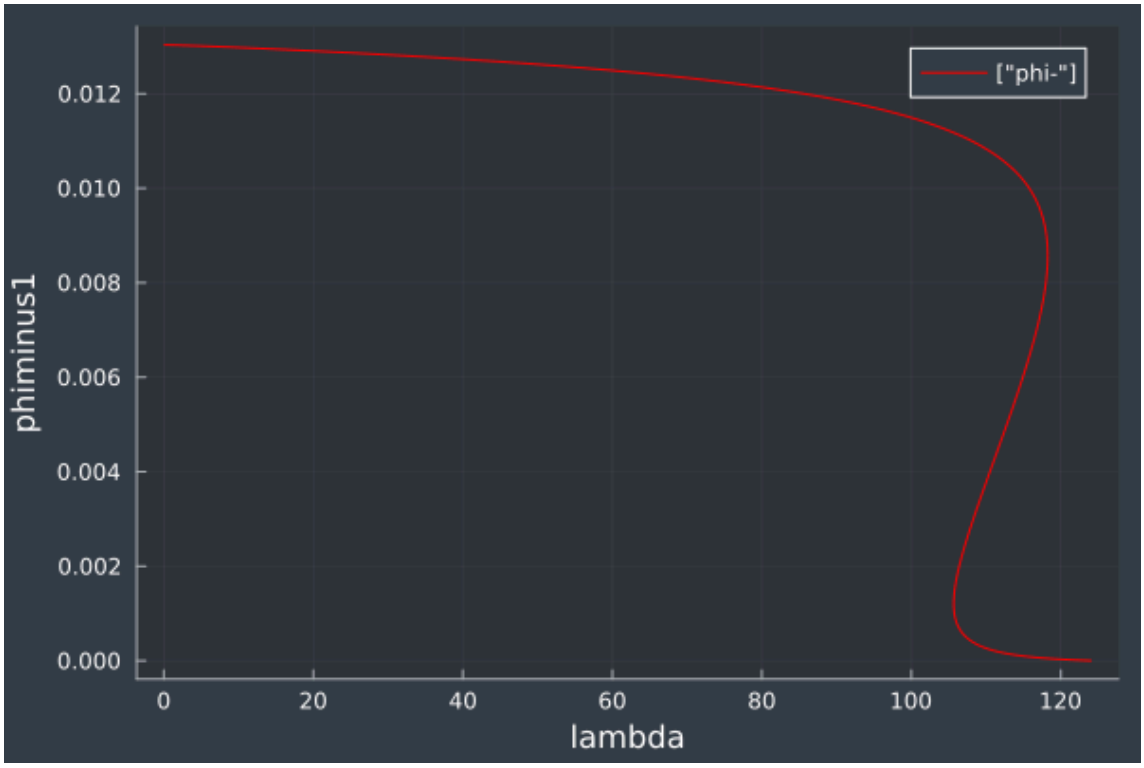
```
plot!(
    qdlen_phiminus1,
    br,
    vars = (:param, :phiplus1),
    branchlabel = ["phi+"],
    legend = :topright,
    color = :green,
)

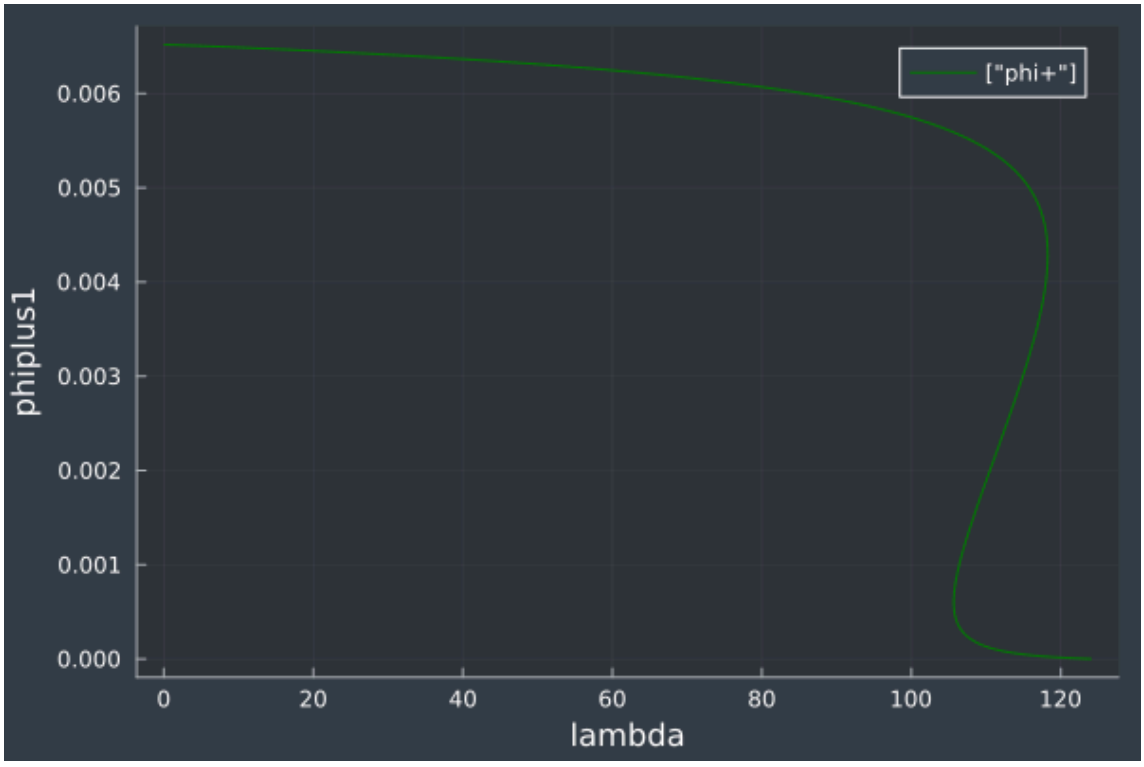
plot!(
    qdlen_phipep1,
    br,
    vars = (:param, :phipep1),
    branchlabel = ["peptide"],
    legend = :topright,
    color = :pink,
)

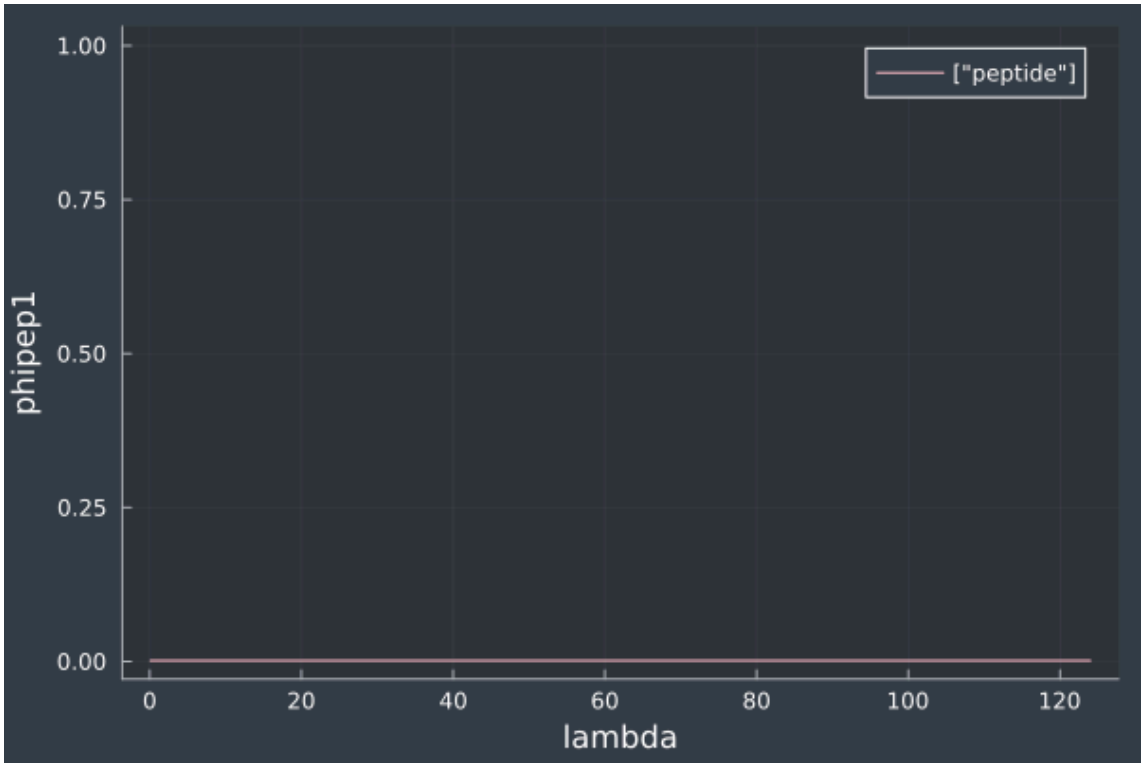
plot!(
    qdlen_phioh1,
    br,
    vars = (:param, :phioh1),
    branchlabel = ["OH-"],
    legend = :topright,
    color = :yellow,
)

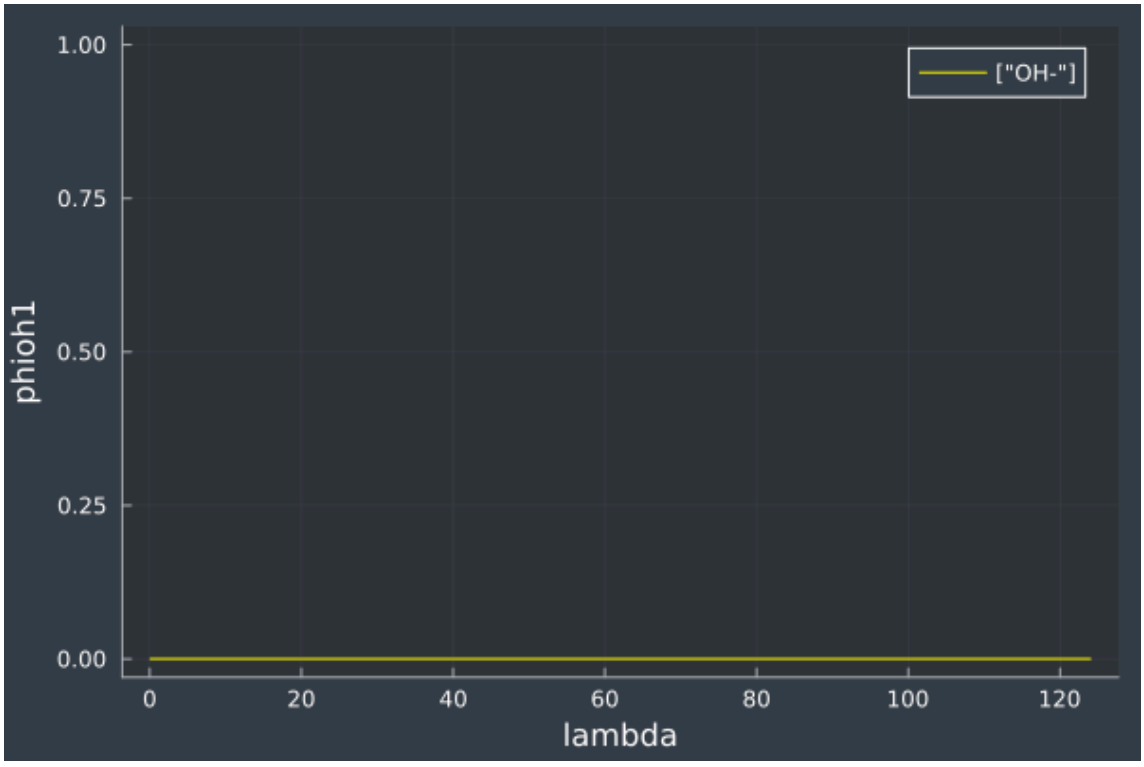
plot!(
    qdlen_phiw1,
    br,
    vars = (:param, :phiw1),
    branchlabel = ["water"],
    legend = :topright,
    color = :lightblue,
)

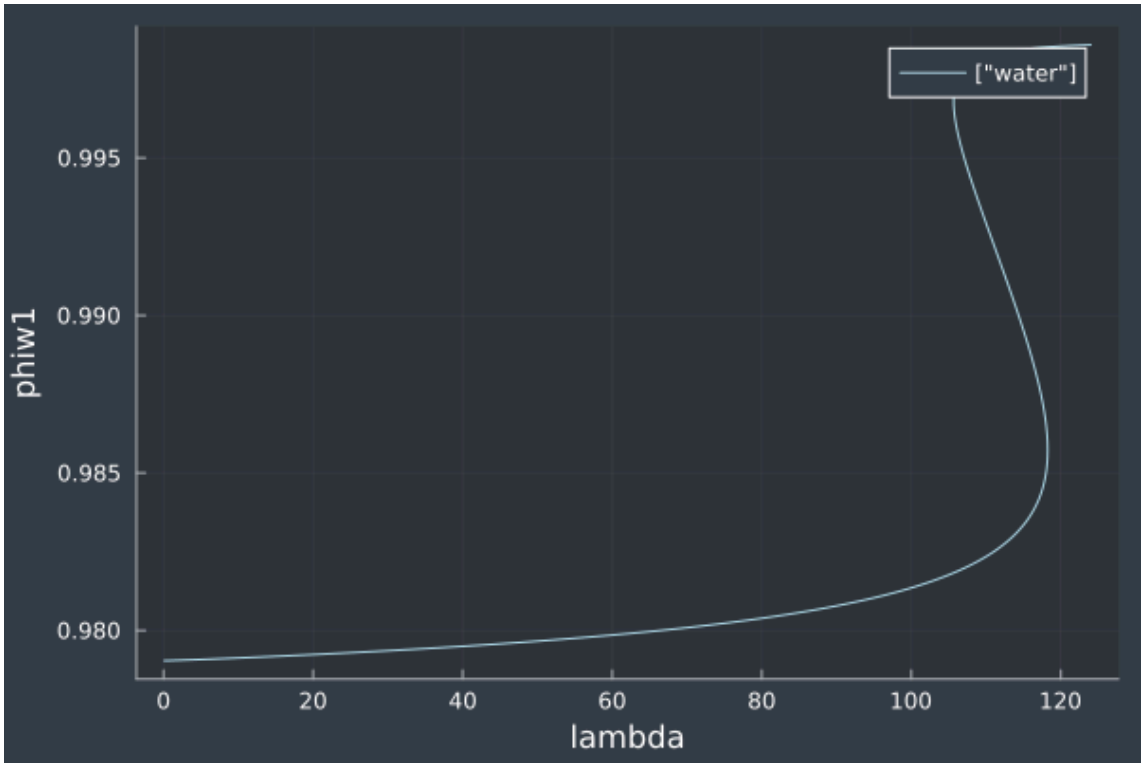
display(qdlen_phiplus1)
display(qdlen_phiminus1)
display(qdlen_phipep1)
display(qdlen_phioh1)
display(qdlen_phiw1)
```


["phi-"]
0.012
0.010
0.008
0.006
0.004
0.002
0.000
phiminus1
0
20
40
60
80
100
120
lambda


["phi+"]
0.006
0.005
0.004
0.003
0.002
0.001
0.000
phiplus1
0
20
40
60
80
100
120
lambda


["peptide"]
phipep1
1.00
0.75
0.50
0.25
0.00
0
20
40
60
80
100
120
lambda


["OH-"]
phioh1
1.00
0.75
0.50
0.25
0.00
0
20
40
60
80
100
120
lambda



In [19]:
```
qdlen_phiplus2 = plot()
qdlen_phiminus2 = plot()
qdlen_phipep2 = plot()
qdlen_phioh2 = plot()
qdlen_phiw2 = plot()

plot!(
    qdlen_phiplus2,
    br,
    vars = (:param, :phiminus2),
    branchlabel = ["phi-"],
    legend = :topright,
    color = :red,
)

plot!(
    qdlen_phiminus2,
    br,
    vars = (:param, :phiplus2),
    branchlabel = ["phi+"],
    legend = :topright,
    color = :green,
)

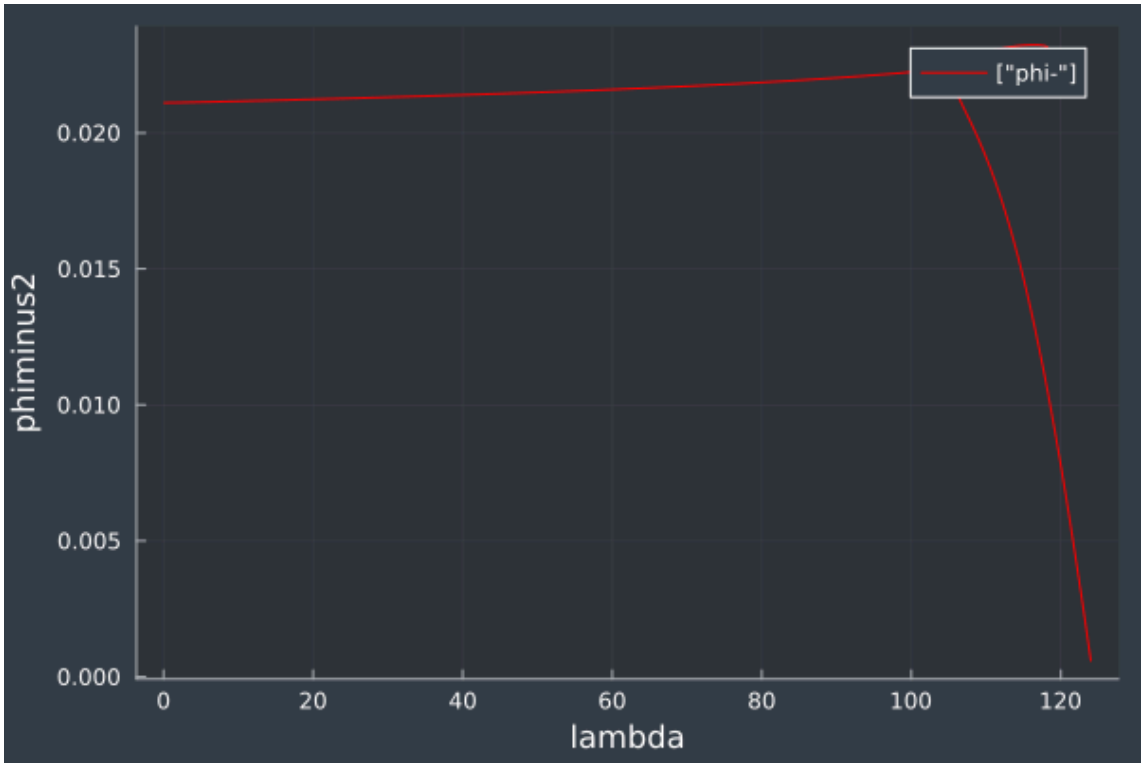
plot!(
    qdlen_phipep2,
    br,
    vars = (:param, :phipep2),
    branchlabel = ["peptide"],
    legend = :topright,
    color = :pink,
)

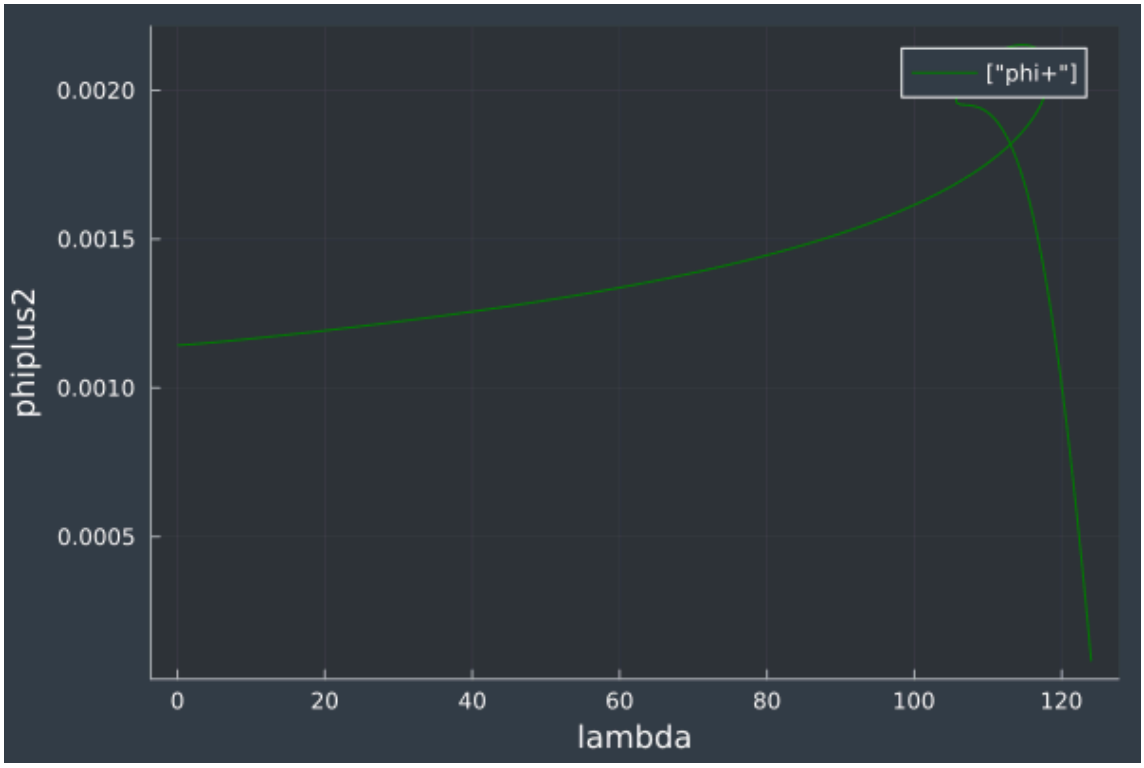
plot!(
    qdlen_phioh2,
    br,
```

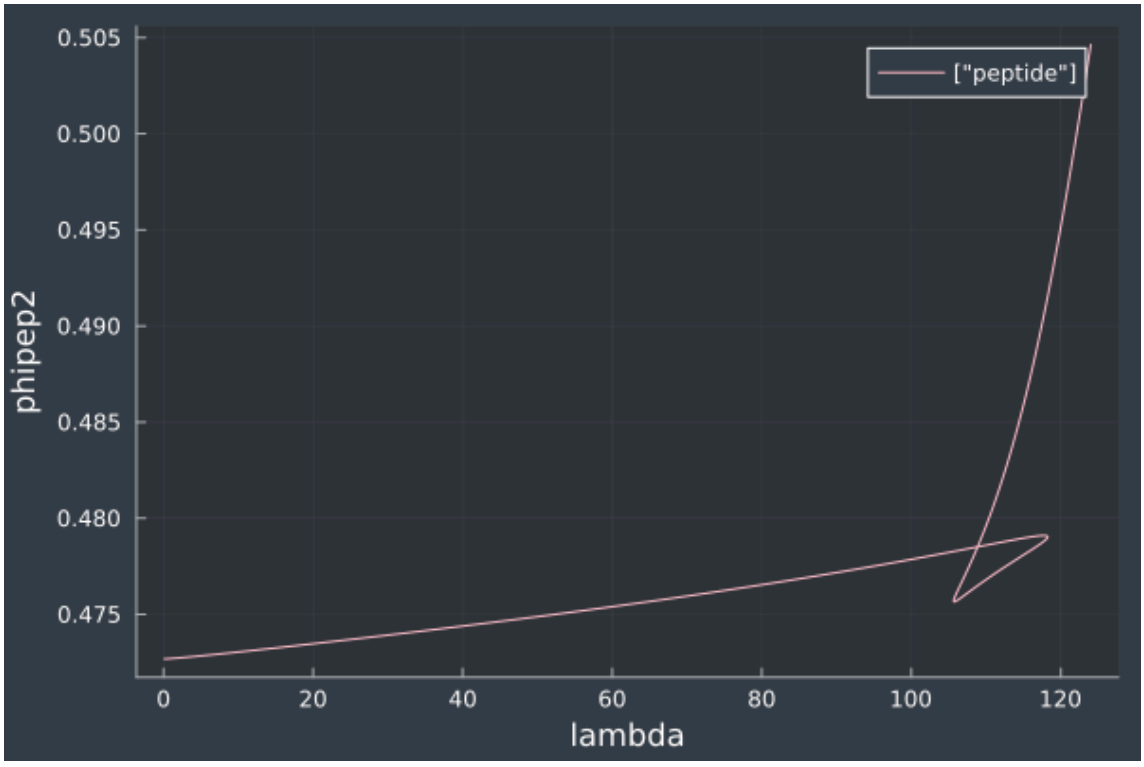
```
    vars = (:param, :phioh2),
    branchlabel = ["OH-"],
    legend = :topright,
    color = :yellow,
)

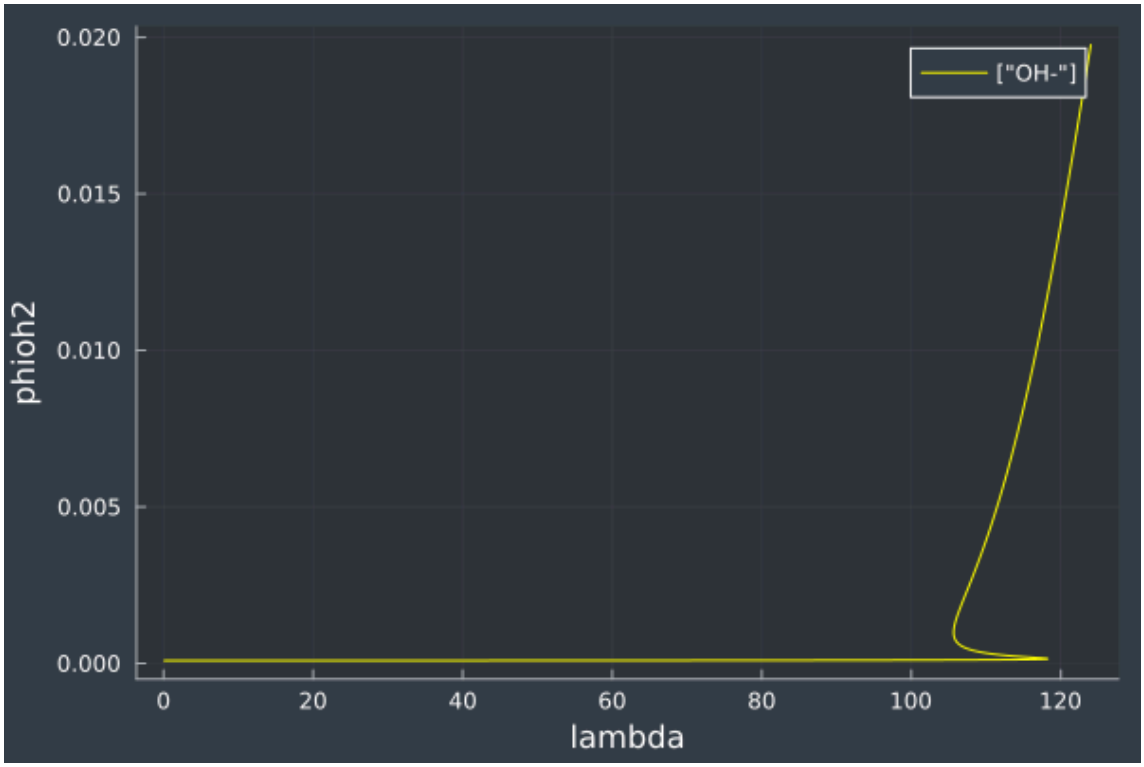
plot!(
    qdlen_phiw2,
    br,
    vars = (:param, :phiw2),
    branchlabel = ["water"],
    legend = :topright,
    color = :lightblue,
)

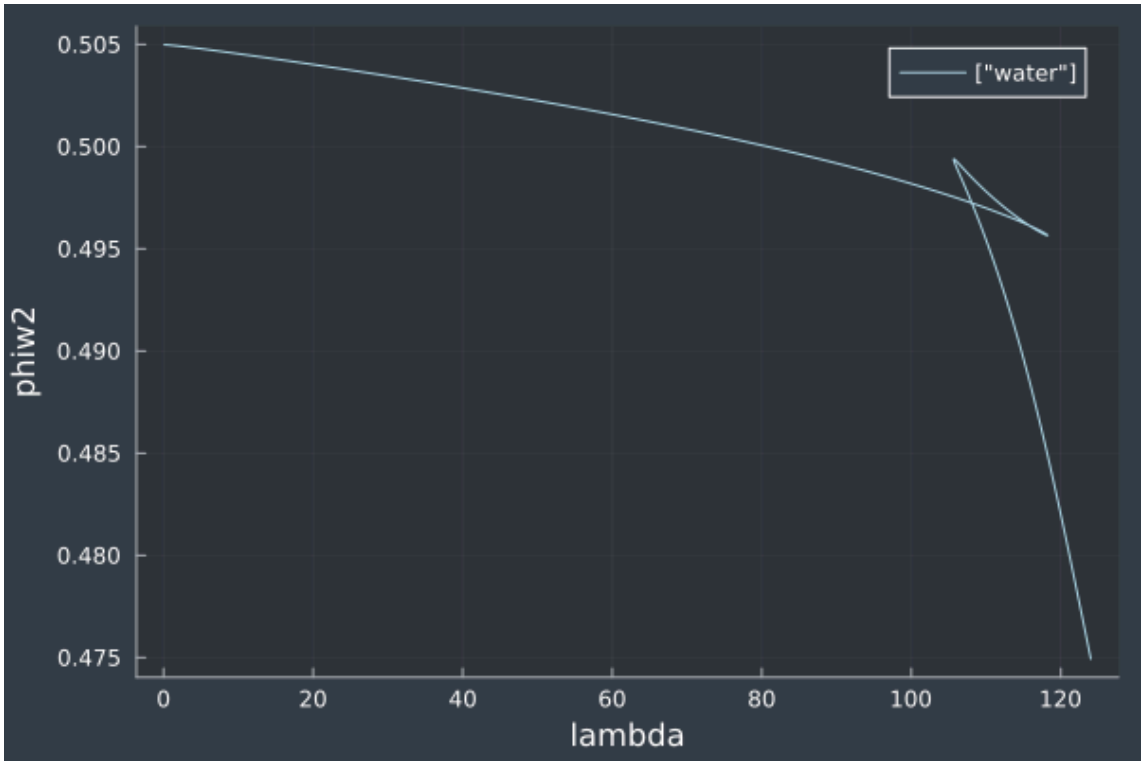
display(qdlen_phiplus2)
display(qdlen_phiminus2)
display(qdlen_phipep2)
display(qdlen_phioh2)
display(qdlen_phiw2)
```




["phi+"]
0.0020
0.0015
0.0010
0.0005
phiplus2
0
20
40
60
80
100
120
lambda


["peptide"]
0.505
0.500
0.495
0.490
0.485
0.480
0.475
phipep2
0
20
40
60
80
100
120
lambda





In [20]:
```
#br.phiw2
typeof(optscontdlen)
fieldnames(typeof(optscontdlen))
```

Out[20]:
```
(:dsmin, :dsmax, :ds, :a, :p_min, :p_max, :max_steps, :newton_options, :
η, :save_to_file, :save_sol_every_step, :nev, :save_eig_every_step, :sav
e_eigenvectors, :plot_every_step, :tol_stability, :detect_fold, :detect_
bifurcation, :dsmin_bisection, :n_inversion, :max_bisection_steps, :tol_
bisection_eigenvalue, :detect_event, :tol_param_bisection_event, :detect
_loop)
```

In [21]:
```
optscontdlen2 = ContinuationPar(
    dsmin = 0.0000001,
    dsmax = 0.00001,
```

```
    ds = 0.000001,
    #        p_max = dd2*1.053499, #6/(1+par.N*par.nuA/2),
    #        p_min = dd2*0.999, #0.00001,
    p_max = par.KM * 500,
    p_min = par.KM * 0.5,
    nev = 0,
    detect_fold = true,
    plot_every_step = 1,
    newton_options = NewtonPar(max_iterations = 500, tol = 1e-12, verbose
    max_steps = 1000000,
    detect_bifurcation = 0,
    tol_stability = 1e-14,
)


#optscontdlen2 = @set @set(optscontdlen.p_min = 1e-8).p_max = 1-1e-8


probdlen2 = BK.re_make(probdlen; lens = (@optic _.KM), u0 = sol.u)

@show probdlen2

@show probdlen2.u0
@show probdlen2.params

br = @time continuation(
    probdlen2,
    alg,
    optscontdlen2;
    plot = false,
    verbosity = 0,
    bothside = true,
)
```

```
probdlen2 = ┌─ Bifurcation problem with uType Vector{Float64}
├─ Inplace: false
├─ Dimension: 17
├─ Symmetric: false
└─ Parameter: KM
probdlen2.u0 = [-11.111350720434567, -10.418188598005115, -6.6025878921893
36, -9.822765447457765, 6.531955189841433, -6.899886898027085, -4.85159264
4655155, -0.019225372606793793, -4.249813262800951, -0.0721200869086027, -
0.9997041172559362, -0.6937810535525439, -10.460277256638797, -7.303167417
9632775, 34.582638252271124, 0.5953988320522502, 0.7014481537145767]
probdlen2.params = (d2 = 0.24654273494601342, chi = 0.8555, N = 7, nuA =
7.142857142857143, lambda = 120.0, KM = 0.0013550135501355014, switch_dh =
1, zplus = 2.0, zpep = [1, 1], p_max = 124.0, p_min = 0.1)
 99.647798 seconds (1.12 G allocations: 145.498 GiB, 21.44% gc time, 1.35%
compilation time)
```

Out[21]:
```
 ┌─ Curve type: EquilibriumCont
 ├─ Number of points: 1355953
 ├─ Type of vectors: Vector{Float64}
 ├─ Parameter KM starts at 0.0006775067750677507, ends at 0.001625696047
9559617
 ├─ Algo: PALC [Bordered]
 └─ Special points:

- #  1, endpoint at KM ≈ +0.00067751,
step =   0
- #  2,     fold at KM ≈ +0.00605875 ∈ (+0.00605875, +0.00605875), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 127240
- #  3,     fold at KM ≈ +0.00068433 ∈ (+0.00068433, +0.00068433), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 267474
- #  4,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1161509
- #  5,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1161510
- #  6,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1163684
- #  7,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1163685
- #  8,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1169035
- #  9,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1169036
- # 10,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1176529
- # 11,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1176530
- # 12,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1177029
- # 13,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1177030
- # 14,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1178341
- # 15,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1178342
- # 16,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1179213
- # 17,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1179214
- # 18,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1184688
- # 19,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1184689
- # 20,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1185330
- # 21,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1185331
- # 22,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1189057
- # 23,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1189058
- # 24,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1190265
- # 25,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1190266
- # 26,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1191301
```

```
- # 27,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1191302
- # 28,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1191579
- # 29,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1191580
- # 30,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1192348
- # 31,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1192349
- # 32,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1193388
- # 33,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1193389
- # 34,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1194252
- # 35,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1194253
- # 36,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1194764
- # 37,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1194765
- # 38,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1194953
- # 39,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1194954
- # 40,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1195966
- # 41,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1195967
- # 42,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1197219
- # 43,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1197220
- # 44,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1199861
- # 45,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1199862
- # 46,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1200514
- # 47,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1200515
- # 48,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1200797
- # 49,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1200798
- # 50,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1201338
- # 51,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1201339
- # 52,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1202712
- # 53,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1202713
- # 54,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1203637
- # 55,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1203638
- # 56,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1203816
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- # 57,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1203817
- # 58,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1204185
- # 59,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1204186
- # 60,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1204547
- # 61,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1204548
- # 62,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1205464
- # 63,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1205465
- # 64,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1206297
- # 65,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1206298
- # 66,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1207859
- # 67,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1207860
- # 68,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1208177
- # 69,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1208178
- # 70,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1209808
- # 71,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1209809
- # 72,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1210029
- # 73,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1210030
- # 74,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1210693
- # 75,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1210694
- # 76,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1211542
- # 77,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1211543
- # 78,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1211955
- # 79,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1211956
- # 80,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1212460
- # 81,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1212461
- # 82,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1212905
- # 83,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1212906
- # 84,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1213666
- # 85,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1213667
- # 86,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1214036
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- # 87,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1214037
- # 88,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1214804
- # 89,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1214805
- # 90,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1215256
- # 91,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1215257
- # 92,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1216103
- # 93,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1216104
- # 94,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1216772
- # 95,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1216773
- # 96,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1217223
- # 97,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1217224
- # 98,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1218492
- # 99,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1218493
- #100,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1219322
- #101,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1219323
- #102,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1219593
- #103,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1219594
- #104,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1220891
- #105,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1220892
- #106,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1221143
- #107,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1221144
- #108,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1221825
- #109,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1221826
- #110,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1222172
- #111,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1222173
- #112,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1223402
- #113,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1223403
- #114,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1224483
- #115,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1224484
- #116,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1224737
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- #117,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1224738
- #118,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1224953
- #119,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1224954
- #120,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1225369
- #121,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1225370
- #122,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1225774
- #123,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1225775
- #124,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1226010
- #125,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1226011
- #126,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1226355
- #127,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1226356
- #128,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1226714
- #129,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1226715
- #130,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1227456
- #131,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1227457
- #132,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1227847
- #133,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1227848
- #134,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1228233
- #135,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1228234
- #136,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1228743
- #137,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1228744
- #138,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1229194
- #139,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1229195
- #140,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1230630
- #141,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1230631
- #142,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1232453
- #143,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1232454
- #144,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1234161
- #145,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1234162
- #146,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1235324
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- #147,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1235325
- #148,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1235511
- #149,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1235512
- #150,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1236078
- #151,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1236079
- #152,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1236459
- #153,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1236460
- #154,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1237064
- #155,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1237065
- #156,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1237527
- #157,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1237528
- #158,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1237742
- #159,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1237743
- #160,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1238375
- #161,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1238376
- #162,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1238593
- #163,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1238594
- #164,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1239224
- #165,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1239225
- #166,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1239428
- #167,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1239429
- #168,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1239627
- #169,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1239628
- #170,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1240786
- #171,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1240787
- #172,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1240967
- #173,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1240968
- #174,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1241335
- #175,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1241336
- #176,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1242081
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- #177,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1242082
- #178,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1242615
- #179,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1242616
- #180,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1242891
- #181,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1242892
- #182,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1243119
- #183,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1243120
- #184,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1243463
- #185,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1243464
- #186,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1243575
- #187,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1243576
- #188,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1243685
- #189,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1243686
- #190,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1245019
- #191,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1245020
- #192,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1245681
- #193,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1245682
- #194,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1246018
- #195,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1246019
- #196,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1247753
- #197,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1247754
- #198,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1248101
- #199,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1248102
- #200,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1249830
- #201,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1249831
- #202,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1250302
- #203,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1250303
- #204,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1250781
- #205,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1250782
- #206,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1251985
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- #207,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1251986
- #208,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1252787
- #209,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1252788
- #210,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1253326
- #211,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1253327
- #212,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1254704
- #213,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1254705
- #214,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1255262
- #215,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1255263
- #216,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1255546
- #217,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1255547
- #218,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1255914
- #219,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1255915
- #220,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1257878
- #221,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1257879
- #222,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1258628
- #223,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1258629
- #224,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1259042
- #225,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1259043
- #226,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1259925
- #227,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1259926
- #228,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1261061
- #229,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1261062
- #230,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1261332
- #231,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1261333
- #232,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1261603
- #233,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1261604
- #234,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1261875
- #235,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1261876
- #236,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1262138
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- #237,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1262139
- #238,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1262402
- #239,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1262403
- #240,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1263439
- #241,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1263440
- #242,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1264362
- #243,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1264363
- #244,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1264511
- #245,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1264512
- #246,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1264656
- #247,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1264657
- #248,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1264949
- #249,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1264950
- #250,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1265235
- #251,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1265236
- #252,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1265380
- #253,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1265381
- #254,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1265668
- #255,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1265669
- #256,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1265810
- #257,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1265811
- #258,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1266236
- #259,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1266237
- #260,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1266523
- #261,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1266524
- #262,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1266804
- #263,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1266805
- #264,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1267327
- #265,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1267328
- #266,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1267781
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- #267,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1267782
- #268,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1268007
- #269,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1268008
- #270,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1268456
- #271,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1268457
- #272,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1268567
- #273,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1268568
- #274,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1268789
- #275,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1268790
- #276,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1268899
- #277,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1268900
- #278,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1269119
- #279,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1269120
- #280,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1269229
- #281,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1269230
- #282,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1269448
- #283,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1269449
- #284,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1269558
- #285,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1269559
- #286,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1269668
- #287,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1269669
- #288,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1269997
- #289,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1269998
- #290,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1270430
- #291,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1270431
- #292,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1270539
- #293,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1270540
- #294,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1270753
- #295,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1270754
- #296,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1271075
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- #297,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1271076
- #298,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1271502
- #299,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1271503
- #300,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1271607
- #301,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1271608
- #302,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1271819
- #303,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1271820
- #304,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1272150
- #305,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1272151
- #306,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1272481
- #307,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1272482
- #308,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1272592
- #309,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1272593
- #310,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1272812
- #311,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1272813
- #312,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1272923
- #313,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1272924
- #314,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1273141
- #315,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1273142
- #316,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1273251
- #317,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1273252
- #318,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1273580
- #319,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1273581
- #320,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1273797
- #321,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1273798
- #322,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1274868
- #323,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1274869
- #324,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1275284
- #325,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1275285
- #326,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1275492
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- #327,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1275493
- #328,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1275799
- #329,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1275800
- #330,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1276005
- #331,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1276006
- #332,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1276312
- #333,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1276313
- #334,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1276516
- #335,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1276517
- #336,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1276730
- #337,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1276731
- #338,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1276925
- #339,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1276926
- #340,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1277305
- #341,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1277306
- #342,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1277554
- #343,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1277555
- #344,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1277803
- #345,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1277804
- #346,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1277926
- #347,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1277927
- #348,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1278053
- #349,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1278054
- #350,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1278303
- #351,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1278304
- #352,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1278554
- #353,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1278555
- #354,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1279698
- #355,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1279699
- #356,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1280463
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- #357,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1280464
- #358,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1280977
- #359,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1280978
- #360,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1281369
- #361,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1281370
- #362,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1281628
- #363,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1281629
- #364,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1281894
- #365,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1281895
- #366,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1282025
- #367,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1282026
- #368,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1282153
- #369,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1282154
- #370,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1282419
- #371,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1282420
- #372,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1282549
- #373,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1282550
- #374,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1282814
- #375,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1282815
- #376,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1282947
- #377,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1282948
- #378,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1283079
- #379,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1283080
- #380,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1283215
- #381,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1283216
- #382,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1283350
- #383,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1283351
- #384,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1283482
- #385,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1283483
- #386,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1283752
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- #387,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1283753
- #388,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1284977
- #389,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1284978
- #390,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1285250
- #391,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1285251
- #392,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1285807
- #393,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1285808
- #394,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1286361
- #395,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1286362
- #396,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1286781
- #397,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1286782
- #398,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1286920
- #399,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1286921
- #400,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1287199
- #401,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1287200
- #402,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1287478
- #403,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1287479
- #404,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1287752
- #405,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1287753
- #406,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1288029
- #407,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1288030
- #408,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1288169
- #409,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1288170
- #410,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1288307
- #411,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1288308
- #412,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1288587
- #413,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1288588
- #414,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1288730
- #415,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1288731
- #416,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1288878
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- #417,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1288879
- #418,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1289160
- #419,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1289161
- #420,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1289441
- #421,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1289442
- #422,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1289581
- #423,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1289582
- #424,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1289865
- #425,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1289866
- #426,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1290719
- #427,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1290720
- #428,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1291008
- #429,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1291009
- #430,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1291440
- #431,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1291441
- #432,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1292022
- #433,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1292023
- #434,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1292459
- #435,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1292460
- #436,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1292605
- #437,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1292606
- #438,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1293043
- #439,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1293044
- #440,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1293338
- #441,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1293339
- #442,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1293636
- #443,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1293637
- #444,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1293938
- #445,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1293939
- #446,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1294235
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- #447,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1294236
- #448,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1294538
- #449,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1294539
- #450,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1294686
- #451,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1294687
- #452,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1294831
- #453,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1294832
- #454,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1295130
- #455,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1295131
- #456,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1295584
- #457,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1295585
- #458,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1295734
- #459,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1295735
- #460,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1295885
- #461,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1295886
- #462,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1296188
- #463,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1296189
- #464,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1296341
- #465,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1296342
- #466,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1296648
- #467,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1296649
- #468,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1297262
- #469,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1297263
- #470,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1300047
- #471,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1300048
- #472,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1300361
- #473,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1300362
- #474,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1300673
- #475,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1300674
- #476,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1300833
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- #477,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1300834
- #478,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1301147
- #479,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1301148
- #480,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1301305
- #481,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1301306
- #482,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1301621
- #483,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1301622
- #484,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1301938
- #485,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1301939
- #486,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1302253
- #487,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1302254
- #488,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1302412
- #489,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1302413
- #490,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1302891
- #491,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1302892
- #492,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1303049
- #493,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1303050
- #494,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1303209
- #495,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1303210
- #496,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1303532
- #497,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1303533
- #498,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1303691
- #499,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1303692
- #500,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1303852
- #501,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1303853
- #502,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1304014
- #503,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1304015
- #504,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1304177
- #505,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1304178
- #506,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1304501
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- #507,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1304502
- #508,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1304663
- #509,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1304664
- #510,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1304826
- #511,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1304827
- #512,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1305156
- #513,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1305157
- #514,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1305318
- #515,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1305319
- #516,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1306804
- #517,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1306805
- #518,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1307636
- #519,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1307637
- #520,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1308135
- #521,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1308136
- #522,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1308472
- #523,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1308473
- #524,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1308641
- #525,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1308642
- #526,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1308978
- #527,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1308979
- #528,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1309314
- #529,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1309315
- #530,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1309651
- #531,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1309652
- #532,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1309989
- #533,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1309990
- #534,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1310554
- #535,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1310555
- #536,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1311002
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- #537,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1311003
- #538,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1311225
- #539,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1311226
- #540,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1311673
- #541,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1311674
- #542,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1311897
- #543,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1311898
- #544,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1312124
- #545,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1312125
- #546,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1312590
- #547,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1312591
- #548,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1312820
- #549,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1312821
- #550,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1313056
- #551,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1313057
- #552,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1313291
- #553,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1313292
- #554,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1313523
- #555,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1313524
- #556,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1313759
- #557,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1313760
- #558,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1314228
- #559,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1314229
- #560,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1314466
- #561,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1314467
- #562,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1314700
- #563,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1314701
- #564,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1314937
- #565,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1314938
- #566,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1315172
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- #567,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1315173
- #568,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1315408
- #569,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1315409
- #570,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1315640
- #571,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1315641
- #572,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1319130
- #573,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1319131
- #574,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1319809
- #575,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1319810
- #576,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1320036
- #577,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1320037
- #578,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1320490
- #579,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1320491
- #580,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1320944
- #581,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1320945
- #582,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1321395
- #583,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1321396
- #584,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1321848
- #585,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1321849
- #586,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1322304
- #587,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1322305
- #588,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1322531
- #589,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1322532
- #590,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1323211
- #591,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1323212
- #592,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1323666
- #593,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1323667
- #594,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1323891
- #595,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1323892
- #596,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1324348
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- #597,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1324349
- #598,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1324576
- #599,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1324577
- #600,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1324803
- #601,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1324804
- #602,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1325257
- #603,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1325258
- #604,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1325487
- #605,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1325488
- #606,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1325715
- #607,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1325716
- #608,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1325945
- #609,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1325946
- #610,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1326173
- #611,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1326174
- #612,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1326628
- #613,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1326629
- #614,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1327088
- #615,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1327089
- #616,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1327315
- #617,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1327316
- #618,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1327774
- #619,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1327775
- #620,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1328005
- #621,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1328006
- #622,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1328460
- #623,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1328461
- #624,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1328691
- #625,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1328692
- #626,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1328919
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- #627,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1328920
- #628,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1329147
- #629,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1329148
- #630,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1329376
- #631,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1329377
- #632,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1329606
- #633,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1329607
- #634,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1329836
- #635,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1329837
- #636,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1330294
- #637,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1330295
- #638,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1332125
- #639,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1332126
- #640,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1334694
- #641,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1334695
- #642,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1335406
- #643,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1335407
- #644,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1335883
- #645,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1335884
- #646,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1336360
- #647,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1336361
- #648,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1336835
- #649,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1336836
- #650,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1337552
- #651,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1337553
- #652,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1338031
- #653,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1338032
- #654,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1338511
- #655,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1338512
- #656,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1338747
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- #657,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1338748
- #658,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1339467
- #659,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1339468
- #660,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1340184
- #661,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1340185
- #662,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1340424
- #663,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1340425
- #664,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1340662
- #665,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1340663
- #666,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1341617
- #667,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1341618
- #668,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1342575
- #669,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1342576
- #670,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1342811
- #671,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1342812
- #672,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1343294
- #673,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1343295
- #674,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1343531
- #675,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1343532
- #676,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1343771
- #677,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1343772
- #678,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1344009
- #679,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1344010
- #680,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1344250
- #681,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1344251
- #682,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1344730
- #683,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1344731
- #684,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1344968
- #685,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1344969
- #686,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1345206
```

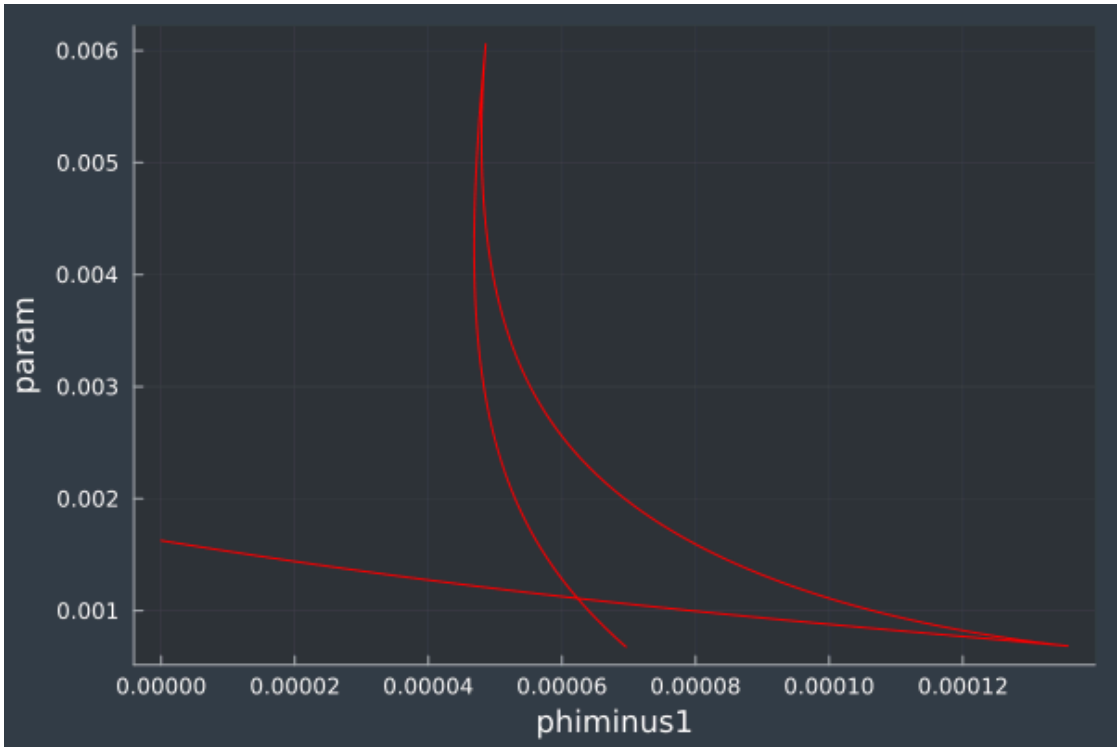
```
- #687,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1345207
- #688,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1345445
- #689,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1345446
- #690,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1345685
- #691,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1345686
- #692,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1345926
- #693,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1345927
- #694,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1346166
- #695,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1346167
- #696,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1346645
- #697,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1346646
- #698,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1346884
- #699,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1346885
- #700,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1347364
- #701,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1347365
- #702,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1347606
- #703,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1347607
- #704,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1347844
- #705,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1347845
- #706,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1348083
- #707,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1348084
- #708,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1348325
- #709,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1348326
- #710,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1348561
- #711,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1348562
- #712,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1348801
- #713,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1348802
- #714,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1349039
- #715,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1349040
- #716,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1349279
```

```
- #717,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1349280
- #718,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1349760
- #719,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1349761
- #720,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1349999
- #721,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1350000
- #722,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1350240
- #723,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1350241
- #724,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1350481
- #725,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1350482
- #726,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1350720
- #727,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1350721
- #728,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1350959
- #729,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1350960
- #730,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1351200
- #731,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1351201
- #732,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1351440
- #733,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1351441
- #734,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1353114
- #735,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1353115
- #736,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1355033
- #737,     fold at KM ≈ +0.00162570 ∈ (+0.00162570, +0.00162570), |δp|=
-1e+00, [    guess], δ = ( 0,  0), step = 1355034
- #738, endpoint at KM ≈ +0.00162570,
step = 1355953
```

In [22]:

```
phiplot = plot(
    br,
    vars = (:phiminus1, :param),
    #    branchlabel = ["br"],
    legend = :topright,
    color = :red,
)
```

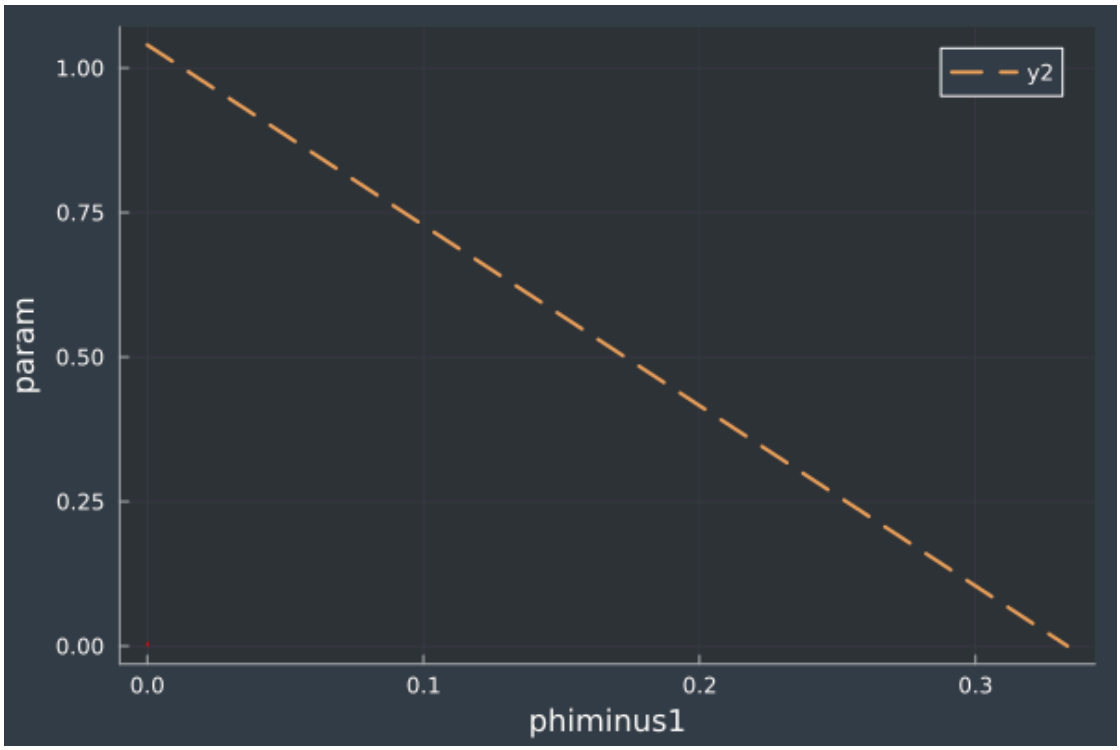
Out[22]:



In [23]:

```
xx = 0:(1/(par.zplus+1))/10:(1/(par.zplus+1))
yy = (1 .- (par.zplus + 1) .* xx) .* (1 + (par.zpep[1] + par.zpep[2]) / (
plot!(phiplot, xx, yy, linewidth = 2, linestyle = :dash)
# collect(xx)
```

Out[23]:

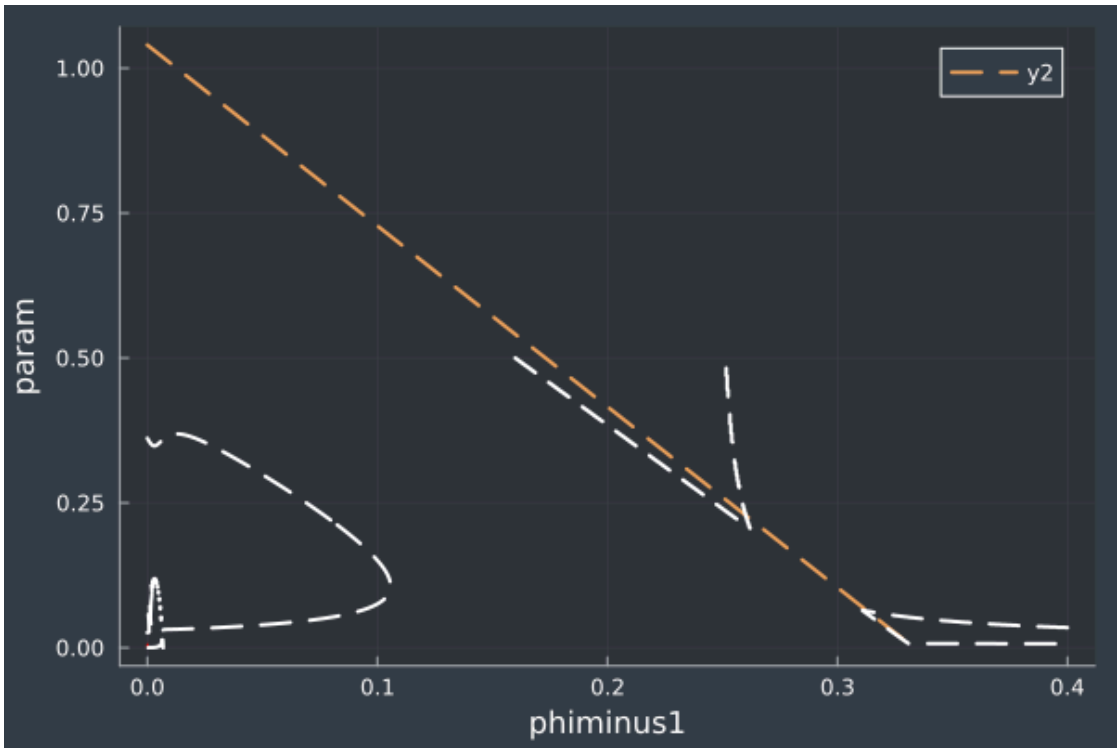


In [24]:

```
contour(
    phiplot,
    phiplus_r,
    phipep_r,
    Z;
    levels = [0],
    linewidth = 2,
    color = :white,
    linestyle = :dash,
```

```
    cbar = false,
)
```

Out[24]:



In [25]:

```
using Printf
import Contour

Zc = permutedims(Z)  # convert from rows=y, cols=x to rows=x, cols=y

open("contour0-souradip-vary-peptide.dat", "w") do io
    println(io, "# contour data")
    println(io, "# columns: x y")
    println(io, "# blank line separates different contour lines")

    cs = Contour.contours(phiplus_r, phipep_r, Zc, [0.0])

    seg = 0
    for cl in Contour.levels(cs)
        for ln in Contour.lines(cl)
            seg += 1
            xs, ys = Contour.coordinates(ln)

            println(io, "# segment ", seg)
            for (x, y) in zip(xs, ys)
                @printf(io, "%.16g %.16g\n", x, y)
            end
            println(io)  # empty line between contour lines
        end
    end
end
```

In [26]:

```
params = br.branch.param
records1 = br.branch.phiplus1
records2 = br.branch.phipep1

# Pair them up in a storeable format
#data_pairs = collect(zip(params, records1, records2));   # Vector{Tuple{
```

```
# Save
#@save "mgcl2_violet_branch_data.jld2" data_pairs;

# Form a 3-column matrix
table = hcat(records1, records2)

# Write as whitespace-separated text
writedlm("binodal-souradip-vary-peptide.dat", table)
```

In [ ]:

In [ ]: